Joshua **Yorko**

Site Reliablity Engineer · Software Engineer

Worcester, MA

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"Site Reliability Engineer and Kubernetes Specialist dedicated to enhancing system resilience and operational excellence."

Skills

Programming Languages	Python, Bash, SQL, JavaScript, TypeScript
Infrastructure & DevOps	Terraform, AWS, GCP, CI/CD (GitHub Actions), Rancher, GitOps
Data Engineering	Python,Apache Iceberg, DuckDB, Spark, Dremio, Nessie,
API Development	FastAPI, Django API Integration & Management, Microservices Architecture
Virtualization & Containerization	EC2, Harvester HCI (SUSE),Proxmox,Docker, Kubernetes,K3s,RKE2,EKS
GitOps	ArgoCD, Flux, Fleet
Full Stack Development	Django, HTML/CSS, JavaScript, React, Next.js, PostgreSQL, Tailwind, Pocketbase
Artificial Intelligence	OpenAI API, AWS BEDROCK, Anthropic, Langchain, Ollama, Llama3

Experience

Gainwell Technologies, Conway, AR

Software Engineer — Platform & Reliability *June 2023 – Present*

- Migrated legacy UiPath automations to Python robots using Robocorp RCC and Sema4AI Action Server, delivering a fully open-source RPA solution under strict cost constraints.
- Designed producer–consumer robots with prebuilt RCC Holotree environments; cut cold-start by ~70% and raised parallel throughput via deterministic env bootstraps.
- Built Playwright-powered web-automation robots and custom ARC (Actions Runner Controller) runner images to enable rapid Kubernetes scale-out for high-concurrency runs.
- Operated multi-account fleets of self-hosted GitHub Actions ARC runner sets to execute Robocorp RPA workloads reliably; implemented file-sharded work items and matrix job sharding (no Redis).
- Designed and executed CI/CD pipelines with GitHub Actions to streamline deployments.
- Orchestrated containerized application delivery on AWS EKS with auto-scaling and HA.
- Developed RESTful services with FastAPI and deployed via Terraform on EKS.
- Automated text extraction with AWS Textract/Lambda into an S3 data lake.
- Managed AWS infrastructure with Terraform (EC2, S3, RDS, ELB, VPC).
- Integrated LLMs via Amazon Bedrock into analytics workflows.
- Built a microservice images API (FastAPI on Kubernetes) to replace deprecated Dental DataClearingHouse endpoints, served via CDN with PostgreSQL and S3.
- Delivered an open data lakehouse on EKS with Dremio + Nessie; transformed IBM3270 data into Apache Iceberg and fed a Python-based adjudication work-distribution algorithm.

Gainwell Technologies, Conway, AR

Senior Data Engineer/Software Engineer *June 2022 - March 2023*

- Architected an enterprise-level data lake utilizing AWS Glue, Amazon S3, PySpark, Hadoop, Python, and PostgreSQL, significantly boosting data processing efficiency.
- Integrated the data lake with Amazon RDS via SQL views for a Django frontend, enabling seamless data access and manipulation.
- Automated complex data retrieval using Python libraries (Requests, aiohttp, asyncio, Selenium, Beautiful Soup), enhancing data acquisition speed and reliability.
- Developed real-time dashboards in PowerBI and custom visualizations using Plotly and Django, delivering insightful data analytics to stakeholders.
- Ensured high availability and scalability of web applications hosted on Amazon EC2.
- Implemented robust security measures, including SSL encryption and firewalls, to safeguard data and applications.
- Proactively monitored applications and infrastructure, minimizing downtime and maximizing operational reliability.

Lambent Spaces, Boston, MA

Implementations Specialist *March 2021 – October 2022*

- Directed software implementations, ensuring customer requirements were met through effective network and system configurations.
- Optimized deployment processes, resulting in heightened efficiency and improved customer satisfaction.
- Fostered collaboration with cross-functional teams to refine deployment workflows and procedures.
- Authored and utilized SQL and Python scripts for automation and efficient problem resolution.

CVS HEALTH, Cumberland, RI

DATA ANALYST (DESKTOP SUPPORT) *May 2019 – March 2021*

- Designed and deployed O365 Apps for Enterprise across 80k devices, leveraging Python and Excel for efficient reporting and achieving a 30% increase in deployment efficiency.
- Created and maintained business-focused reports using Dash/Plotly, SharePoint, and Excel, driving informed decision-making processes.
- Conducted extensive data normalization from SCCM and Active Directory, ensuring data accuracy and consistency.
- Played a pivotal role in Windows 10 Enterprise Migration, focusing on rigorous testing and quality assurance of developed reports.

SPENCER TECHNOLOGIES, Medway, MA

CLIENT ONBOARDING SPECIALIST *Jan 2017 – Apr 2019*

- Implemented the Smartsheet Request Management System, resulting in annual cost savings of $16k.
- Successfully onboarded new business ventures worth over $15 million, enhancing company growth and market presence.
- Developed and executed project plans for multiple IT projects, leading teams to successful and timely client launches.

Education

Indiana State University

Terre Haute, IN

B.A. IN ENGLISH *Aug 2006 – June 2008*

Robocorp Migration & Orchestration

Personal / Enterprise Automation Initiatives

ROBOCORP MIGRATION & ORCHESTRATION *June 2023 – Present*

- Maintains a **personal fork of RCC (v18.5 OSS)** with targeted patches to speed environment provisioning and orchestration.
- Designed high-throughput producer/consumer robots with prebuilt **RCC Holotree** environments, cutting robot cold-start by ~75% and boosting parallel throughput.
- Built Playwright-based web automation robots with optimized startup, reducing end-to-end cycle time significantly.
- Operated multi-account fleets of self-hosted GitHub Actions ARC runner sets to execute Robocorp workloads reliably at scale.
- Implemented file-adapter work items, file sharding, and matrix job sharding (no Redis), improving utilization and simplifying retries.

Cloud & Containerized Robotics

KUBERNETES & DEVCONTAINERS *June 2023 – Present*

- Built custom **ARC (Actions Runner Controller)** Docker images with prebuilt Holotree envs for rapid scale-out on Kubernetes, improving deployment speed by ~60%.
- Standardized reproducible environments with **Dev Containers** across local and cloud infrastructure.
- Integrated Docker with **RCC Remote** for seamless remote execution and debugging of robots.
- Scaled via self-hosted ARC runner sets; adopted file adapters + file sharding; designed matrix job sharding to spread load across runners.

Open Source Contributions

AUTOMATION TOOLS & PLATFORMS *June 2023 – Present*

- **RCC (personal fork, v18.5 OSS):** maintain a private fork (`github.com/joshyorko/rcc`) with targeted patches to improve Holotree provisioning and CI usage in resource-constrained environments; build reproducible binaries internally.
- **control-room (ongoing):** Robocorp-inspired orchestration platform prototype for managing robot lifecycles and workload distribution.
- **fetch-repos-bot**: workflow automation, repo discovery/sync, and CI/CD helpers (`github.com/joshyorko/fetch-repos-bot`).
- **room-of-requirement**: reproducible Dev Container / Holotree environment scaffolding (`github.com/joshyorko/room-of-requirement`).
- **rccremote-docker**: remote execution + Docker integration for RCC (`github.com/yorko-io/rccremote-docker`).
- **arc-runner-custom-docker**: purpose-built repo to launch ARC runner scale sets on Kubernetes with custom container images for RPA (Playwright + RCC Holotree) (`github.com/yorko-io/arc-runner-custom-docker`).
- Stack demonstrates open-source, budget-friendly RPA patterns (RCC + ARC + Playwright + Dev Containers) and multi-account automation approaches.